

Cryptography in Cloud Security

(information in unreadable form)

Learn :

x How cloud providers use encryption (Azure Key Vault).

Cloud providers use **Azure Key Vault** to manages/protects :

- **Secret** : tokens, passwords, certificates, and API keys etc
- **Key** : Azure Key vault (used for encryption)
- **Certificates** : TLS/SSL

x Transport Layer Security (TLS) & SSL certificates.

Both are cryptographic protocols that secures communications over the network/internet.

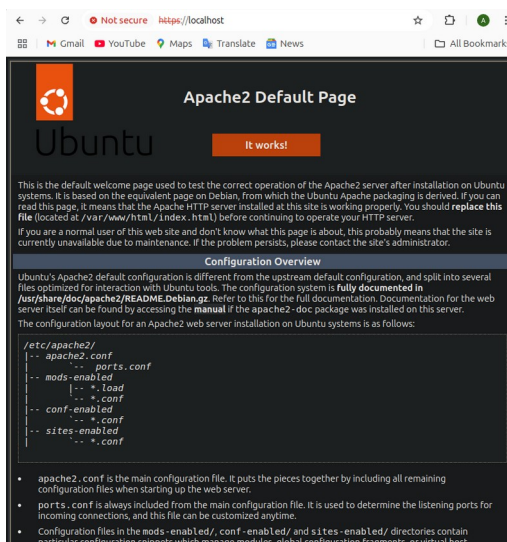
Feature	SSL (Secure Sockets Layer)	TLS (Transport Layer Security)
Introduction	Developed by Netscape in 1995	Introduced as an improved version of SSL in 1999
Security	Less secure, vulnerable to attacks	More secure with advanced cryptographic algorithms
Versions	SSL 2.0, SSL 3.0 (both deprecated)	TLS 1.0, 1.1 (deprecated), TLS 1.2, TLS 1.3 (latest)
Performance	Slower, consumes more resources	Faster, optimized handshake, reduced latency
Cipher Suites	Supports weaker algorithms (e.g., MD5, SHA-1)	Uses stronger encryption (e.g., AES, SHA-256, ChaCha20)
Handshake Process	More complex and slower	Faster handshake, reduced round trips
Usage Today	No longer recommended	Standard encryption for secure internet communication

Task :

- x **Generate a self-signed SSL certificate and install it on a local web server.**

```
aditi@aditi-IdeaPad-3-15IAU7:~$ openssl x509 -in /etc/ssl/certs/mycert.crt -text -noout
Certificate:
    Data:
        Version: 3 (0x2)
        Serial Number:
            45:27:45:57:87:32:7c:1f:be:47:ca:3c:51:cb:09:77:4e:44:d6:26
        Signature Algorithm: sha256WithRSAEncryption
        Issuer: C = IN, ST = Madhya Pradesh, L = Bhopal, O = CS, OU = CS,
        CN = Aditi Gupta, emailAddress = esha3392@gmail.com
        Validity
            Not Before: Mar  4 14:35:32 2025 GMT
            Not After : Mar  4 14:35:32 2026 GMT
        Subject: C = IN, ST = Madhya Pradesh, L = Bhopal, O = CS, OU = CS
        , CN = Aditi Gupta, emailAddress = esha3392@gmail.com
        Subject Public Key Info:
            Public Key Algorithm: rsaEncryption
            Public-Key: (2048 bit)
            Modulus:
                00:b0:16:c0:21:ee:89:c5:28:e7:8e:43:55:2f:ea:
                f6:45:ea:f1:72:9d:f4:ab:f9:ac:e5:c4:32:6d:55:
                12:dd:32:bf:de:d5:9b:d7:66:1b:b7:4e:d1:66:53:
                0d:2e:e5:52:3a:e0:36:ec:60:6d:d9:6b:f1:f7:a3:
                6a:9e:6e:23:0b:d6:d0:23:0f:f1:f3:72:33:ab:d6:
```

<https://localhost/> : shows default Apache Page (SSL is Configured Correctly)



x Enable encryption for a cloud storage bucket (Azure Blob Storage)

Azure Blob storage provides encryption at rest by default by using Azure Storage Service Encryption (option 1)

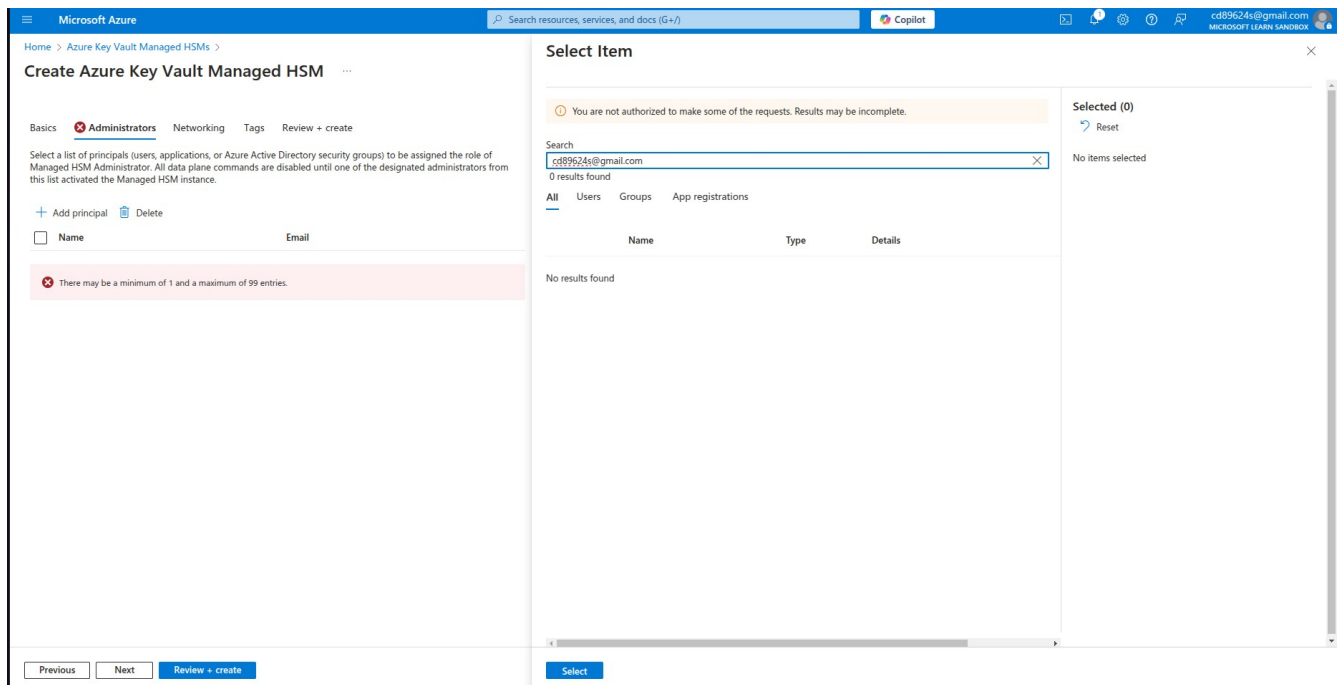
The screenshot shows the Microsoft Azure portal interface. The left sidebar displays the 'Storage accounts' section with a list of accounts, including 'cloudshell1557129751'. The main content area is titled 'cloudshell1557129751 | Encryption'. It features a search bar and a list of navigation options: Overview, Activity log, Tags, Diagnose and solve problems, Access Control (IAM), Data migration, Storage browser, Storage Mover, Partner solutions, Data storage, Security + networking, Networking, Front Door and CDN, Access keys, Shared access signature, Encryption (selected), Microsoft Defender for Cloud, Data management, Settings, Monitoring, Monitoring (classic), Automation, and Help. The 'Encryption' section is expanded, showing 'Encryption selection' with 'Enable support for customer-managed keys' set to 'Disabled'. Under 'Encryption type', 'Microsoft-managed keys' is selected with a radio button, and 'Customer-managed keys' is unselected. A 'Save' button is visible at the bottom right of the settings area.

Encryption type

- ☒ Microsoft-managed keys
- ☐ Customer-managed keys

Enable Customer Managed Key (option 2)

it give you full control over encryption keys, enhancing security, compliance, and access management



I tried but, I have limited permissions (Microsoft sandbox)

~Task 3 Week 5 (I)

~March 5, 2025